form; however, Euler's relation can be used to change them into a cosine and sine form if needed.

3. Analysis and Synthesis

Each transform has an analysis equation (also called the forward transform) and a synthesis equation (also called the inverse transform). The analysis equations describe how to calculate each value in the frequency domain based on all of the values in the time domain. The synthesis equations describe how to calculate each value in the time domain based on all of the values in the frequency domain.

4. Time Domain Notation

Continuous time domain signals are called $x(t)$, while discrete time domain signals are called $x[n]$. For the complex transforms, these signals are complex. For the real transforms, these signals are real. All of the time domain signals extend from minus infinity to positive infinity. However, if the time domain is periodic, we are only concerned with a single cycle, because the rest is redundant. The variables, T and N, denote the periods of continuous and discrete signals in the time domain, respectively.

5. Frequency Domain Notation

Continuous frequency domain signals are called $X(\omega)$ if they are complex, and $ReX(\omega)$ & $ImX(\omega)$ if they are real. Discrete frequency domain signals are called $X[k]$ if they are complex, and $ReX[k]$ & $ImX[k]$ if they are real. The complex transforms have negative frequencies that extend from minus infinity to zero, and positive frequencies that extend from zero to positive infinity. The real transforms only use positive frequencies. If the frequency domain is periodic, we are only concerned with a single cycle, because the rest is redundant. For continuous frequency domains, the independent variable, ω, makes one complete period from $-\pi$ to π. In the discrete case, we use the period where k runs from 0 to N-1

6. The Analysis Equations

The analysis equations operate by *correlation*, i.e., multiplying the time domain signal by a sinusoid and integrating (continuous time domain) or summing (discrete time domain) over the appropriate time domain section. If the time domain signal is aperiodic, the appropriate section is from minus infinity to positive infinity. If the time domain signal is periodic, the appropriate section is over any one complete period. The equations shown here are written with the integration (or summation) over the period: 0 to T (or 0 to N-1). However, any other complete period would give identical results, i.e., $-T$ to 0, $-T/2$ to $T/2$, etc.

7. The Synthesis Equations

The synthesis equations describe how an individual value in the time domain is calculated from all the points in the frequency domain. This is done by multiplying the frequency domain by a sinusoid, and integrating (continuous frequency domain) or summing (discrete frequency domain) over the appropriate frequency domain section. If the frequency domain is complex and aperiodic, the appropriate section is negative infinity to positive infinity. If the